

The fate of Amazon precipitation after massive deforestation

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Abstract

Deforestation in the Amazon has been predicted to cause irreversible forest dieback, primarily due to significant reductions in mean precipitation driven by decreased evapotranspiration. However, these results are based on conventional climate models that use convective parameterizations and/or limited boundaries, leading to precipitation with high sensitivity to evapotranspiration and restricted large-scale interactions. To overcome these limitations, we use a storm-resolving global climate model run at a 5 km grid spacing to simulate the response of precipitation to complete deforestation over the Amazon basin. We find no significant change in annual precipitation and a distinct spatial pattern of precipitation change compared to previous studies. This suggests that the Amazon may be resilient to deforestation when assessed using conventional metrics of stability, as annual mean precipitation. However, we observe an increase in the intensity of extreme hourly precipitation at both ends – no rain and violent rain – indicating potential risks to the forest ecosystem. We identify key mechanisms that differentiate the storm-resolving model from previous conventional models, highlighting the missed basic physical processes in previous studies, which distorted their response to precipitation from Amazon deforestation. Additionally, we underscore the importance of considering shortterm precipitation, often masked by long-term averages, to more accurately evaluate the impacts of deforestation.

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